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DUST CYCLONE DESIGN

P. A. Funk, S. E. Hughs, G. A. Holt

ABSTRACT. Dust cyclones are a cost effective means of treating process air released by cotton gins. Modifications to the currently recommended cyclone design were tested to improve collection efficiency. Factors normally considered uncontrolled (dust loading and relative humidity) were included in a mixed-level fractional factorial experimental design. Pre-weighed gin trash was metered into inlet air. Exhaust air was passed through a filter that was weighed to determine dust loading and collection efficiency. Square and angled inlet modifications lowered cyclone performance. Neutral air tube and expansion chamber modifications were insignificant. Pressure drop increased linearly with increasing air entrance velocity while collection efficiency decreased slightly. Reducing entrance velocity may save energy without adversely affecting performance.

Keywords. Dust Cyclones, Modeling, Cotton ginning.

The USDA-ARS cotton ginning research laboratories at Lubbock, Texas, and Mesilla Park, New Mexico experiment with dust cyclones in hopes of improving collection efficiency. Dust cyclones are a cost effective way for cotton gins to clean conveying air before it is released to the environment. Other methods of dust collection such as rotary drum filters, bag houses, and spray systems are more expensive to build and operate. A typical 30 bale per hour cotton gin might require as much as 113 m³/s (240,000 cfm) total air for transporting materials between successive stages of the ginning process. Depending on crop condition and source, some of that conveying air may carry as much as 100 g/m³ (44 grains/ft³) total suspended particulate (TSP). The Environmental Protection Agency (1987) limits allowable dust emission of particles having an aerodynamic diameter of less than 10 microns (PM₁₀) to 150 µg/m³ (66 × 10⁻⁶ grains/ft³). Local jurisdictions may have more stringent requirements. Cyclones alone usually suffice to remove TSP and PM₁₀. If air quality regulations become more restrictive, more efficient cyclones will be needed for gins to continue to economically comply with air quality permitting requirements.

ANTECEDENTS

Chemical industry cyclone separators are occasionally designed with inlets at various angles other than perpendicular to the cyclone axis. For this reason positive and negative angled entrance ducts were examined to determine their potential for improving dust cyclone performance. Previous USDA cyclone research had indicated performance improvements when using an expansion chamber at the bottom of a cyclone employing a tapered air outlet, though not with regular air outlets (Holt et al., 1999). Further experimentation was desired to confirm the potential of this modification. The Clifford tube and the expansion chamber were each expected to stabilize the bottom of the vortex, reducing particle re-entrainment at the trash exit. The importance of minimizing re-circulation of lint and fine particles near the bottom of the cone has been identified previously (Baker et al., 1996) in reducing emissions and wear.

INITIAL MODELING

Computational Fluid Dynamics software (AEA Technology plc, 1999) was used to predict airflow streamlines in a dust cyclone. The modeled space was the cyclone barrel and cone volumes. The boundary was drawn at the bottom of the vortex finder because the software could not accept a separate volume within an active volume.

The resulting graphic outputs enabled visualization of airflow rather than particle flow. In the plan view of cyclone streamlines (fig. 1) air makes one revolution in the barrel of the cyclone, collides with incoming air, and makes an abrupt change in direction near the entrance. This forces the dust-laden air to make a tight turn inward and downward. The fact that the air is leaving the dust behind at that point appeared to be more important to cyclone efficiency than the fact that the dust is reintroduced into incoming air. This alternate approach to understanding cyclone dynamics was thought to possibly explain the results of an earlier experiment, which attempted to prevent particle reintroduction.

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Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U. S. Department of Agriculture.

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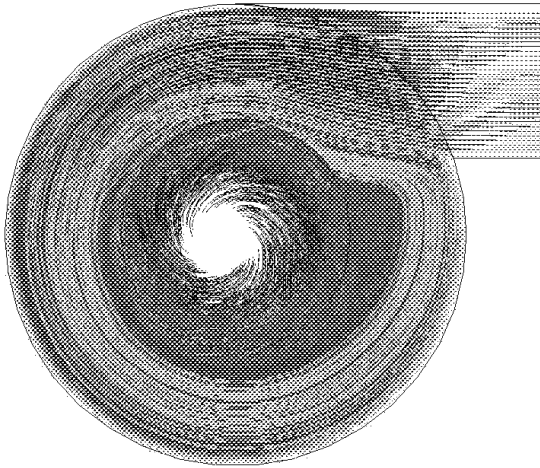


Figure 1. Plan view of cyclone streamlines generated by CFX computational fluid dynamics numerical modeling software.

The inlet helix was placed inside the cyclone cylinder and spiraled downward from a point at the top of the air inlet extending one revolution around the cyclone and ended at the bottom of the air inlet. [The] inlet helix reduced cyclone back pressure. Unfortunately, [it] also reduced collection efficiency. About three times as much trash material was lost from cyclones using [this] device as compared to an unmodified cyclone (Baker and Stedronski, 1967).

Adding the helix had prevented the re-entrainment of particles crossing the incoming air stream, but paradoxically it had also reduced the collection efficiency. This led to the square inlet portion of the experiment.

Another attempt to model cyclone performance was based on the Leith and Licht model (Leith and Licht, 1972; referenced by Cooper and Alley, 1994). Results from that algorithm showed efficiency increasing with air entrance velocity indefinitely, whereas experience has shown that there is a practical upper limit (efficiency decreases at the upper end, as in fig. 2). This observation led to the differing entrance velocity portion of the experiment.

EXPERIMENTAL DESIGN

Figure 3 illustrates the general design of the experiment. Six barrels were constructed having square or rectangular inlets angled -10° , 0° , and 10° from horizontal (fig. 3a). Two cone sections were constructed, one with an expansion

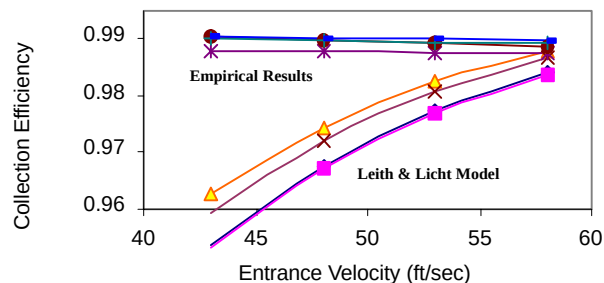


Figure 2. Leith and Licht model compared to empirical results for four cyclone designs.

chamber and one with a plain cone (both with a terminal diameter of $D/3$). To these cone sections were fitted sealed buckets or buckets with a Clifford tube extending half way up the center of the cone (fig. 3b). Various possible combinations of barrel and cone were realized by bolting together the interchangeable barrels and cones as called for by a computer-generated randomized fractional mixed-level experiment design (SAS Institute, 1997). Velocity and dust loading were varied over three and two levels, respectively. A total of 27 observations were made because of a sub-test examining relative humidity influence, as indicated by table 1.

MATERIALS AND METHODS

A preweighed 60×60 -cm (24×24 -in.) Hi-vol fiberglass filter captured the particles that were not removed from the air stream by the cyclone. A flat-blade fan operating at constant speed drew air through the filter. A variable speed fan was used to set the air velocity entering the cyclone. A standard pitot tube was used to set the air flow before each run to one of three predetermined velocity pressure levels corresponding to actual cyclone inlet air velocities of 13, 15.5, or 18 m/s (2560, 3050, or 3540 ft/min). Trials that included the Clifford tube required a two-step set up. The cyclone was run with clean air to determine Clifford tube suction and flow. A hood connecting the cyclone exhaust to

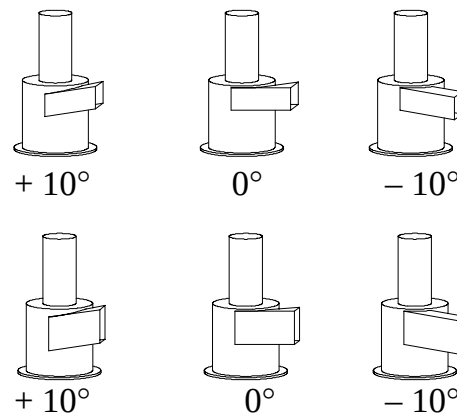


Figure 3a. Square and rectangular inlets at three different angles.

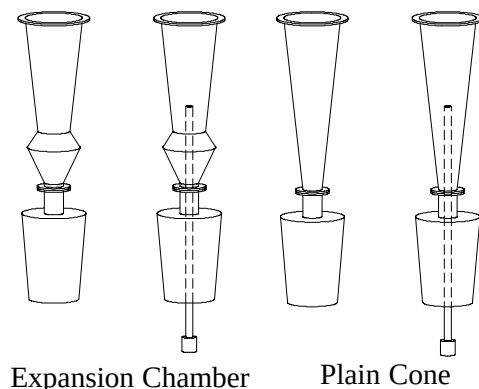


Figure 3b. Expansion chamber and plain cone sections used in empirical trials.

Table 1. Experimental design with cyclone configuration, entrance velocity and dust loading.

Lot	Shape	Angle (°)	Cone	Bucket	Velocity (m/s)	Load (g/m ³)
1	Square	0	Expansion	C-tube	18	16
2	Rectangle	-10	Plain	C-tube	15.5	16
3	Rectangle	10	Plain	Standard	18	8
4	Square	10	Plain	C-tube	15.5	8
5	Rectangle	0	Expansion	C-tube	13	8
6	Rectangle	-10	Expansion	C-tube	18	16
7	Rectangle	0	Plain	Standard	18	8
8	Square	0	Plain	Standard	13	16
9	Square	10	Plain	C-tube	13	16
10	Square	10	Expansion	C-tube	18	16
11	Square	0	Expansion	Standard	15.5	16
12	Rectangle	10	Expansion	Standard	13	8
13	Rectangle	10	Expansion	Standard	15.5	16
14	Square	-10	Expansion	Standard	15.5	8
15	Square	-10	Expansion	Standard	18	8
16	Rectangle	0	Plain	C-tube	15.5	8
17	Rectangle	-10	Plain	Standard	13	16
18	Square	-10	Expansion	C-tube	13	8
0	Rectangle	-10	Plain	C-tube	18	16
h1	Rectangle	0	Expansion	Standard	15.5	16
h2	Rectangle	0	Expansion	Standard	15.5	8
h3	Rectangle	0	Expansion	Standard	15.5	16
h4	Rectangle	0	Expansion	Standard	15.5	8
d1	Rectangle	0	Expansion	Standard	15.5	16
d2	Rectangle	0	Expansion	Standard	15.5	8
d3	Rectangle	0	Expansion	Standard	15.5	16
d4	Rectangle	0	Expansion	Standard	15.5	8

the filter was added and the previously determined airflow was reestablished while the filter fan was running. A small plastic cone inserted into the Clifford tube entrance bell regulated airflow much like a needle valve (fig. 4).

The trash catch bucket was weighed and attached to the bottom of the cyclone being tested. A conveyor belt loaded with gin trash proportional to one of two dust loading rates, 8 or 16 g/m³ (3.5 or 7 grains/ ft³), metered a steady stream of particles (size distribution determined by Hughs and Baker, 1998) into the inlet of the variable speed fan for 120 s. The cyclone catch bucket and filter were weighed again to determine the proportion of material captured by the cyclone and lost in its exhaust.

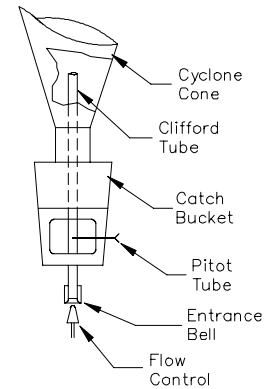


Figure 4. Detail showing Clifford tube and its flow control mechanism.

Pressures and air temperatures were recorded with an HP34970 data logger. The test apparatus is depicted in figure 5. Data analysis was performed using JMP software (SAS Institute, Inc, 1997). The probability that each parameter made a significant contribution to cyclone efficiency was tested at the 5% level of confidence.

RESULTS

DESIGN PARAMETERS

Table 2 presents average efficiencies for each of the experimental cyclone modifications and the corresponding control. Two of the four design modifications, the expansion chamber and the Clifford tube, resulted in cyclone efficiencies that were not significantly different from the control. The expansion chamber results agree with results published previously by Holt et al. (1999). The Clifford tube results were not expected; that modification made no difference in cyclone efficiency despite introducing an additional 20% airflow (by volume) into the heart of the vortex. However, if dust loading of exhaust air is measured, it was reduced proportionately.

The square inlet resulted in lower cyclone efficiency, which also was not expected. The theory that a more abrupt airflow direction change would improve performance did not hold in this test, possibly because the inlet partially overlapped the exhaust duct (vortex finder) near the center.

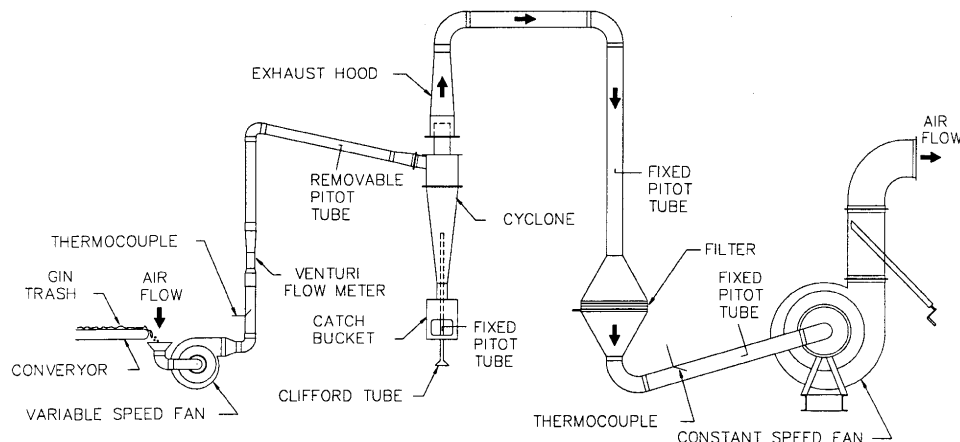


Figure 5. Test Apparatus used to perform these experiments.

Table 2. Results from screening trials.

Parameter	Control and Modification	Average Efficiency	Probability > F
Entrance shape	Rectangle	0.9928 b	0.0014
	Square	0.9915 a	
Entrance angle	−10°	0.9928 b	<0.0001
	0°	0.9931 b	
	+10°	0.9906 a	
Entrance velocity	13 m/s	0.9931 b	0.0006
	15.5 m/s	0.9927 b	
	18 m/s	0.9910 a	
Dust loading	8 g/m ³	0.9917 a	0.0003
	16 g/m ³	0.9930 b	
Cone configuration	Plain and expansion chamber	0.9921 a	n.s.
		0.9919 a	
Collection bucket	Standard and Clifford tube	0.9918 a	n.s.
		0.9922 a	
Relative humidity	Dry (20%) and	0.9932 a	n.s.
	Wet (80%)	0.9933 a	

A 2D2D-type cyclone inlet still results in the best performance.

The angle of the entrance did impact cyclone efficiency in a statistically significant way. Introducing dust-burdened air into a cyclone at an upward angle of 10° such that it strikes the top of the cyclone reduced collection efficiency by a quarter of one percent, from 0.9931 to 0.9906. Introducing the airflow downward did not reduce efficiency substantially from the efficiency realized with a horizontal inlet duct. In absolute terms the difference was 0.025%.

UNCONTROLLED PARAMETERS

Loading rate was also statistically significant. As shown by many researchers previously, a lightly loaded cyclone is not as efficient as one heavily loaded. While it has a lower total emission in g/m³, a higher proportion of the dust entering the cyclone escapes to the atmosphere.

Theoretically, the efficiency of a cyclone should increase as air velocity and inertia increase. But turbulence caused by minor surface imperfections also increases, and dust particles that were at the wall can be thrown back into the air stream. In this study, the reduction in efficiency attributed to turbulence exceeded the gains attributed to particle inertia. Efficiency decreased with increasing entrance velocity. Further testing on full-scale cyclones with open trash outlets is needed to verify these findings. There is some concern that eliminating airflow at the trash outlet may have masked some of the velocity effects. There also is concern that minor variances in fabrication may have a large impact on performance. Figure 6 presents a curve fitting cyclone efficiency to velocity. The whisker plots indicate one standard deviation.

DISCUSSION

Figure 7 is a plot of the cyclone operating pressure at each entrance velocity. The whisker plots indicate one standard deviation. Cyclone pressure drop is important because fan motors pushing air through cyclones consume a significant

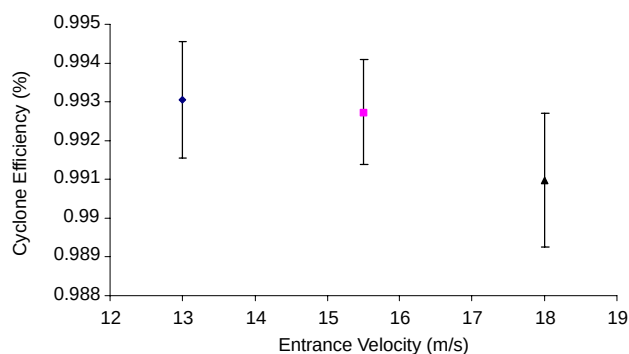


Figure 6. Cyclone efficiency as a function of entrance velocity.

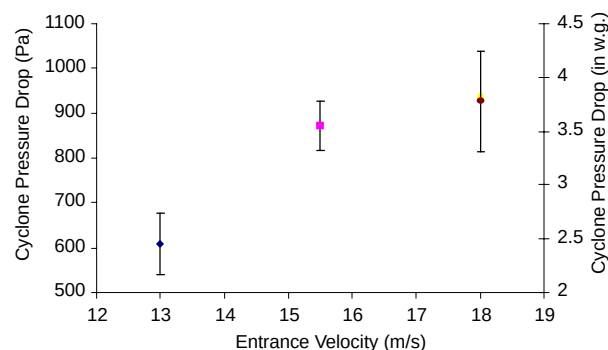


Figure 7. Cyclone pressure drop as a function of entrance velocity.

portion of the energy used by a modern cotton gin. For the scale model cyclones tested, there appears to be an advantage to running the cyclones at a lower speed. This result must be confirmed with full-scale cyclones in an actual ginning situation. Lowering the cyclone entrance velocity merits further investigation because it would not significantly increase emissions, and it possibly could reduce energy consumption.

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